

TE Curve Verification Pipeline Multi-Model Curve Comparison Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report compares representative TE Curve Verification Pipeline TE curves by overlaying multiple model predictions on the same original measured curve. The plots are intended to show whether each model tracks the local harmonic oscillations rather than only the broad mean trend.

Scope

- each comparison image contains four deterministic held-out test curves;
- forward comparisons are shown on forward curves only;
- backward comparisons are shown on backward curves only;
- Wave 1 screening keeps the three strongest family-best models by Curve MAE [deg] within each direction;
- Original Curve uses the same visual weight as predictions and a dark-gray color for balanced comparison.

Metrics Summary

Forward Reference Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_original_best_Fw	rcim_original	Fw	0.002769	0.002951	6.250
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.003014	0.003204	6.819

Forward Wave 1 Family Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
feedforward_fw	wave1_current_registry	Fw	0.003404	0.003855	7.551
harmonic_regression_fw	wave1_current_registry	Fw	0.003230	0.003494	7.185
periodic_mlp_fw	wave1_current_registry	Fw	0.003254	0.003553	7.232
residual_harmonic_mlp_fw	wave1_current_registry	Fw	0.003273	0.003563	7.266
tree_fw	wave1_current_registry	Fw	0.003053	0.003395	6.731
periodic_mlp_harmonic_fw	wave1_periodic_mlp_harmonic_campaign	Fw	0.003254	0.003553	7.232

Backward Reference Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.005027	0.005212	11.860

Backward Wave 1 Family Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
feedforward_bw	wave1_current_registry	Bw	0.003586	0.004023	7.832
harmonic_regression_bw	wave1_current_registry	Bw	0.003678	0.004012	8.058
periodic_mlp_bw	wave1_current_registry	Bw	0.003574	0.004006	7.807
residual_harmonic_mlp_bw	wave1_current_registry	Bw	0.003536	0.003874	7.728
tree_bw	wave1_current_registry	Bw	0.003258	0.003651	7.051
periodic_mlp_harmonic_bw	wave1_periodic_mlp_harmonic_campaign	Bw	0.003583	0.003925	7.875

Forward Wave 2.1 Temporal Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.003603	0.004031	8.028
gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.003330	0.003762	7.378
lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.003366	0.003800	7.450
periodic_temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.003335	0.003708	7.404
periodic_gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.003186	0.003438	7.077
periodic_lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.003266	0.003550	7.258

Backward Wave 2.1 Temporal Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.003742	0.004166	8.184
gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.003626	0.004082	7.907
lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.003555	0.003985	7.767
periodic_temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.003628	0.003987	7.979
periodic_gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.002392	0.002639	5.466
periodic_lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.002625	0.002877	6.013

Forward Wave 2.3 Residual Harmonic Temporal Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
residual_harmonic_gru_sequence_sparse_rcim_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.003194	0.003499	7.083
residual_harmonic_gru_sequence_dense240_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.006983	0.009275	15.722
residual_harmonic_gru_sequence_dense360_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007869	0.010574	17.740
residual_harmonic_lstm_sequence_sparse_rcim_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.003229	0.003533	7.164
residual_harmonic_lstm_sequence_dense240_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007042	0.009370	15.868
residual_harmonic_lstm_sequence_dense360_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007731	0.010235	17.430

Backward Wave 2.3 Residual Harmonic Temporal Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
residual_harmonic_gru_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.003502	0.003857	7.654
residual_harmonic_gru_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.008984	0.012987	20.358
residual_harmonic_gru_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.009370	0.013165	21.267
residual_harmonic_lstm_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.003440	0.003793	7.510
residual_harmonic_lstm_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.007367	0.009945	16.660
residual_harmonic_lstm_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.010268	0.014769	23.355

Forward Wave 3.1 Offset-Aware Probe Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_1_sequential_residual_offset_probe_fw	wave3_1_offset_aware_probe_registry	Fw	0.003377	0.003799	7.487

Backward Wave 3.1 Offset-Aware Probe Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_1_sequential_residual_offset_probe_bw	wave3_1_offset_aware_probe_registry	Bw	0.003636	0.004065	7.952

Forward Wave 3.2 Harmonic-Offset Probe Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_2_clean_sequential_residual_offset_fw	wave3_2_harmonic_offset_probe_registry	Fw	0.003439	0.003870	7.632
wave3_2_harmonic_residual_offset_fw	wave3_2_harmonic_offset_probe_registry	Fw	0.002850	0.003108	6.286

Backward Wave 3.2 Harmonic-Offset Probe Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_2_clean_sequential_residual_offset_bw	wave3_2_harmonic_offset_probe_registry	Bw	0.003541	0.003971	7.732
wave3_2_harmonic_residual_offset_bw	wave3_2_harmonic_offset_probe_registry	Bw	0.003331	0.003671	7.261

Forward Wave 3.3 Curve-Aware Training Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_3_curve_aware_pointwise_control_fw	wave3_3_curve_aware_training_registry	Fw	0.003362	0.003612	7.474
wave3_3_raw_centered_shape_curve_aware_fw	wave3_3_curve_aware_training_registry	Fw	0.003174	0.003429	7.047
wave3_3_raw_offset_curve_aware_fw	wave3_3_curve_aware_training_registry	Fw	0.003269	0.003588	7.268
wave3_3_full_curve_composite_fw	wave3_3_curve_aware_training_registry	Fw	0.003251	0.003515	7.209

Backward Wave 3.3 Curve-Aware Training Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_3_curve_aware_pointwise_control_bw	wave3_3_curve_aware_training_registry	Bw	0.003436	0.003761	7.538
wave3_3_raw_centered_shape_curve_aware_bw	wave3_3_curve_aware_training_registry	Bw	0.003465	0.003790	7.582
wave3_3_raw_offset_curve_aware_bw	wave3_3_curve_aware_training_registry	Bw	0.003469	0.003799	7.608
wave3_3_full_curve_composite_bw	wave3_3_curve_aware_training_registry	Bw	0.003510	0.003897	7.683

Forward Wave 4.1 Robust-Loss Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_1_mae_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.003134	0.003382	6.956
wave4_1_smooth_l1_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.003300	0.003545	7.342
wave4_1_log_cosh_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.003344	0.003595	7.427

Backward Wave 4.1 Robust-Loss Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_1_mae_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.003433	0.003750	7.506
wave4_1_smooth_l1_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.003078	0.003403	6.676
wave4_1_log_cosh_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.003486	0.003811	7.628

Forward Wave 4.2 Quantile Probabilistic Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_2_quantile_p10_p50_p90_fw	wave4_2_probabilistic_registry	Fw	0.003276	0.003545	7.279
wave4_2_gaussian_nll_fw	wave4_2_probabilistic_registry	Fw	0.003156	0.003415	7.008

Backward Wave 4.2 Quantile Probabilistic Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_2_quantile_p10_p50_p90_bw	wave4_2_probabilistic_registry	Bw	0.002935	0.003250	6.307
wave4_2_gaussian_nll_bw	wave4_2_probabilistic_registry	Bw	0.003001	0.003303	6.488

Forward Wave 4.3 Mixture Density Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_3_mixture_density_k2_fw	wave4_3_mixture_density_registry	Fw	0.003329	0.003593	7.388
wave4_3_mixture_density_k3_fw	wave4_3_mixture_density_registry	Fw	0.003226	0.003487	7.164

Backward Wave 4.3 Mixture Density Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_3_mixture_density_k2_bw	wave4_3_mixture_density_registry	Bw	0.002668	0.002947	5.880
wave4_3_mixture_density_k3_bw	wave4_3_mixture_density_registry	Bw	0.002730	0.003009	6.049

Forward Wave 4.4 Latent State Hysteresis Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_4_gru_latent_offset_residual_fw	wave4_4_latent_state_hysteresis_registry	Fw	0.003549	0.003996	7.873
wave4_4_causal_tcn_latent_offset_residual_fw	wave4_4_latent_state_hysteresis_registry	Fw	0.003476	0.003939	7.717

Backward Wave 4.4 Latent State Hysteresis Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_4_gru_latent_offset_residual_bw	wave4_4_latent_state_hysteresis_registry	Bw	0.003542	0.003984	7.736
wave4_4_causal_tcn_latent_offset_residual_bw	wave4_4_latent_state_hysteresis_registry	Bw	0.003624	0.004098	7.903

Forward Wave 5.1 Harmonic Prior Residual Registry Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave5_1_harmonic_prior_pointwise_control_fw	wave5_1_harmonic_prior_residual_registry	Fw	0.003374	0.003655	7.501
wave5_1_harmonic_prior_smooth_l1_structured_fw	wave5_1_harmonic_prior_residual_registry	Fw	0.003514	0.003768	7.812

Backward Wave 5.1 Harmonic Prior Residual Registry Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave5_1_harmonic_prior_pointwise_control_bw	wave5_1_harmonic_prior_residual_registry	Bw	0.003360	0.003677	7.363
wave5_1_harmonic_prior_smooth_l1_structured_bw	wave5_1_harmonic_prior_residual_registry	Bw	0.003431	0.003739	7.523

Forward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.003014	0.003204	6.819
tree_fw	wave1_current_registry	Fw	0.003053	0.003395	6.731
harmonic_regression_fw	wave1_current_registry	Fw	0.003230	0.003494	7.185
periodic_mlp_fw	wave1_current_registry	Fw	0.003254	0.003553	7.232

Backward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.005027	0.005212	11.860
tree_bw	wave1_current_registry	Bw	0.003258	0.003651	7.051
residual_harmonic_mlp_bw	wave1_current_registry	Bw	0.003536	0.003874	7.728
periodic_mlp_bw	wave1_current_registry	Bw	0.003574	0.004006	7.807

Forward Reference Tree And Wave 2.1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.003014	0.003204	6.819
tree_fw	wave1_current_registry	Fw	0.003053	0.003395	6.731
temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.003603	0.004031	8.028
gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.003330	0.003762	7.378
lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.003366	0.003800	7.450
periodic_temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.003335	0.003708	7.404
periodic_gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.003186	0.003438	7.077
periodic_lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.003266	0.003550	7.258

Backward Reference Tree And Wave 2.1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.005027	0.005212	11.860
tree_bw	wave1_current_registry	Bw	0.003258	0.003651	7.051
temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.003742	0.004166	8.184
gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.003626	0.004082	7.907
lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.003555	0.003985	7.767
periodic_temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.003628	0.003987	7.979
periodic_gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.002392	0.002639	5.466
periodic_lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.002625	0.002877	6.013

Forward Reference Tree And Wave 2.3 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.003014	0.003204	6.819
tree_fw	wave1_current_registry	Fw	0.003053	0.003395	6.731
residual_harmonic_gru_sequence_sparse_rcim_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.003194	0.003499	7.083
residual_harmonic_gru_sequence_dense240_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.006983	0.009275	15.722
residual_harmonic_gru_sequence_dense360_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007869	0.010574	17.740
residual_harmonic_lstm_sequence_sparse_rcim_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.003229	0.003533	7.164
residual_harmonic_lstm_sequence_dense240_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007042	0.009370	15.868
residual_harmonic_lstm_sequence_dense360_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007731	0.010235	17.430

Backward Reference Tree And Wave 2.3 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.005027	0.005212	11.860
tree_bw	wave1_current_registry	Bw	0.003258	0.003651	7.051
residual_harmonic_gru_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.003502	0.003857	7.654
residual_harmonic_gru_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.008984	0.012987	20.358
residual_harmonic_gru_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.009370	0.013165	21.267
residual_harmonic_lstm_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.003440	0.003793	7.510
residual_harmonic_lstm_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.007367	0.009945	16.660
residual_harmonic_lstm_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.010268	0.014769	23.355

Forward Reference Tree And Wave 3.1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.003014	0.003204	6.819
tree_fw	wave1_current_registry	Fw	0.003053	0.003395	6.731
wave3_1_sequential_residual_offset_probe_fw	wave3_1_offset_aware_probe_registry	Fw	0.003377	0.003799	7.487

Backward Reference Tree And Wave 3.1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.005027	0.005212	11.860
tree_bw	wave1_current_registry	Bw	0.003258	0.003651	7.051
wave3_1_sequential_residual_offset_probe_bw	wave3_1_offset_aware_probe_registry	Bw	0.003636	0.004065	7.952

Forward Reference Tree And Wave 3.3 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.003014	0.003204	6.819
tree_fw	wave1_current_registry	Fw	0.003053	0.003395	6.731
wave3_3_curve_aware_pointwise_control_fw	wave3_3_curve_aware_training_registry	Fw	0.003362	0.003612	7.474
wave3_3_raw_centered_shape_curve_aware_fw	wave3_3_curve_aware_training_registry	Fw	0.003174	0.003429	7.047
wave3_3_raw_offset_curve_aware_fw	wave3_3_curve_aware_training_registry	Fw	0.003269	0.003588	7.268
wave3_3_full_curve_composite_fw	wave3_3_curve_aware_training_registry	Fw	0.003251	0.003515	7.209

Backward Reference Tree And Wave 3.3 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.005027	0.005212	11.860
tree_bw	wave1_current_registry	Bw	0.003258	0.003651	7.051
wave3_3_curve_aware_pointwise_control_bw	wave3_3_curve_aware_training_registry	Bw	0.003436	0.003761	7.538
wave3_3_raw_centered_shape_curve_aware_bw	wave3_3_curve_aware_training_registry	Bw	0.003465	0.003790	7.582
wave3_3_raw_offset_curve_aware_bw	wave3_3_curve_aware_training_registry	Bw	0.003469	0.003799	7.608
wave3_3_full_curve_composite_bw	wave3_3_curve_aware_training_registry	Bw	0.003510	0.003897	7.683

Forward Reference Tree And Wave 4 series Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.003014	0.003204	6.819
tree_fw	wave1_current_registry	Fw	0.003053	0.003395	6.731
wave4_1_mae_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.003134	0.003382	6.956
wave4_1_smooth_l1_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.003300	0.003545	7.342
wave4_1_log_cosh_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.003344	0.003595	7.427

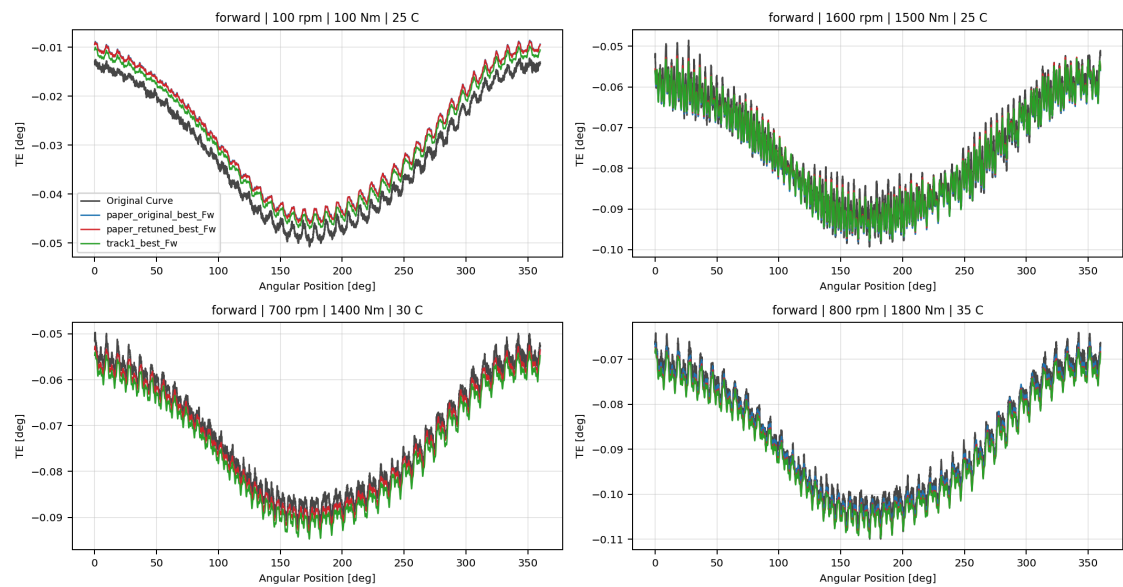
Backward Reference Tree And Wave 4 series Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.005027	0.005212	11.860
tree_bw	wave1_current_registry	Bw	0.003258	0.003651	7.051
wave4_1_mae_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.003433	0.003750	7.506
wave4_1_smooth_l1_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.003078	0.003403	6.676
wave4_1_log_cosh_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.003486	0.003811	7.628

Comparison Gallery - Forward Reference Model Overlay

Included models: `paper_original_best_Fw` , `paper_retuned_best_Fw` , `rcim_model_bank_reproduction_best_fw` .

Forward Reference Model Overlay

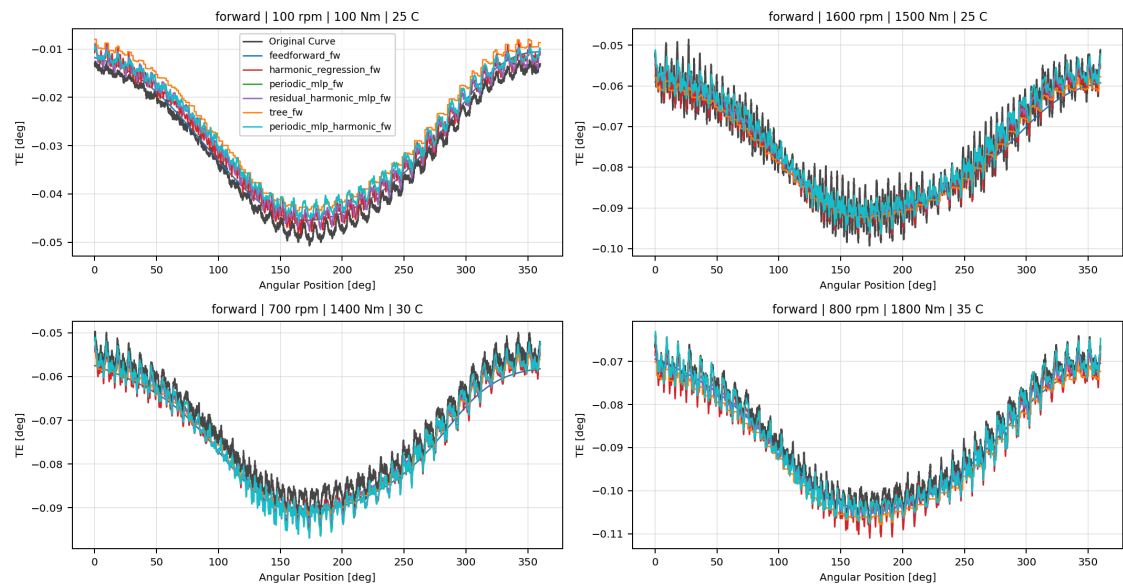


Forward Reference Model Overlay curve-verification comparison

Comparison Gallery - Forward Wave 1 Family Model Overlay

Included models: `feedforward_fw` , `harmonic_regression_fw` , `periodic_mlp_fw` , `residual_harmonic_mlp_fw` , `tree_fw` , `periodic_mlp_harmonic_fw` .

Forward Wave 1 Family Model Overlay

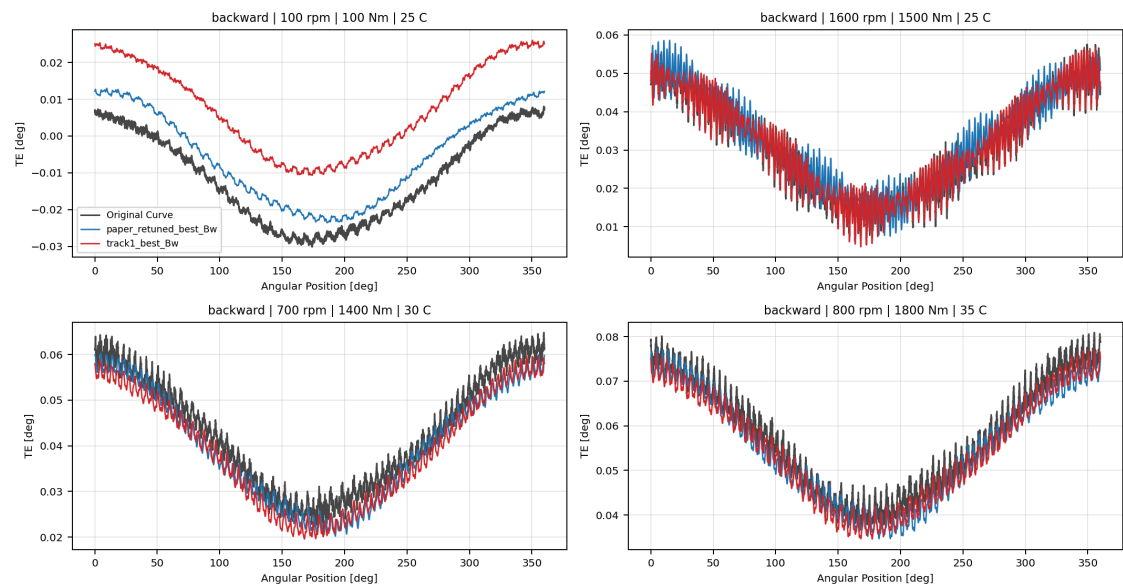


Forward Wave 1 Family Model Overlay curve-verification comparison

Comparison Gallery - Backward Reference Model Overlay

Included models: `paper_retuned_best_Bw` , `rcim_model_bank_reproduction_best_bw` .

Backward Reference Model Overlay

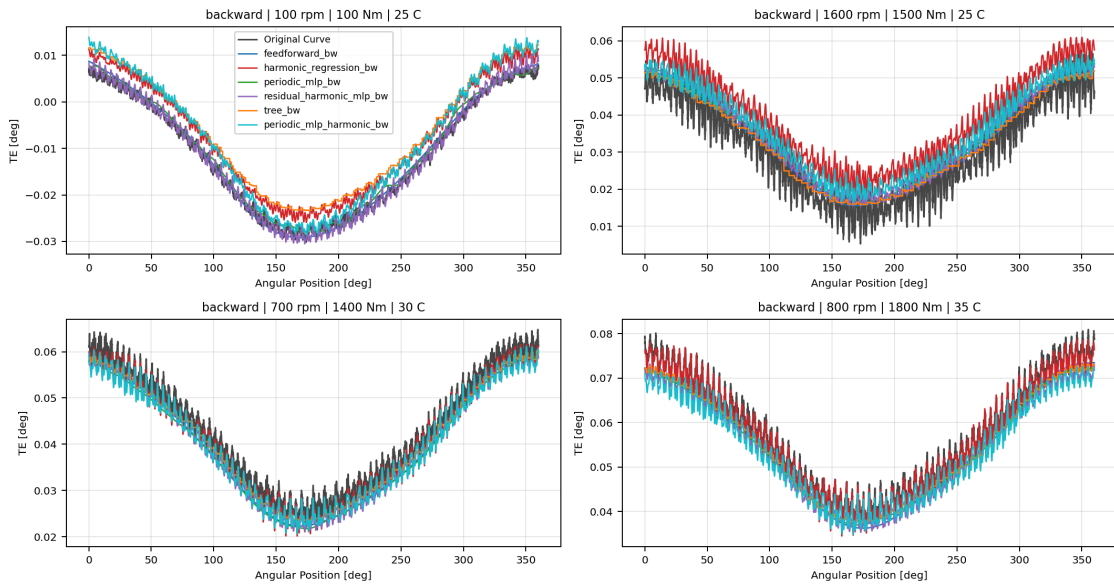


Backward Reference Model Overlay curve-verification comparison

Comparison Gallery - Backward Wave 1 Family Model Overlay

Included models: `feedforward_bw` , `harmonic_regression_bw` , `periodic_mlp_bw` , `residual_harmonic_mlp_bw` , `tree_bw` , `periodic_mlp_harmonic_bw` .

Backward Wave 1 Family Model Overlay

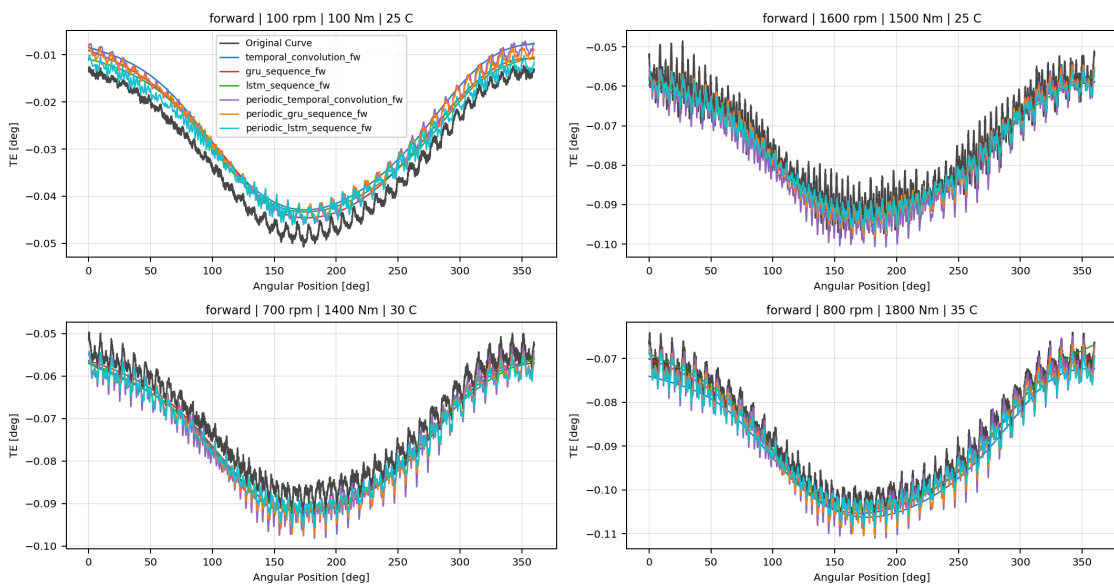


Backward Wave 1 Family Model Overlay curve-verification comparison

Comparison Gallery - Forward Wave 2.1 Temporal Model Overlay

Included models: `temporal_convolution_fw` , `gru_sequence_fw` , `lstm_sequence_fw` , `periodic_temporal_convolution_fw` , `periodic_gru_sequence_fw` , `periodic_lstm_sequence_fw` .

Forward Wave 2 Temporal Model Overlay

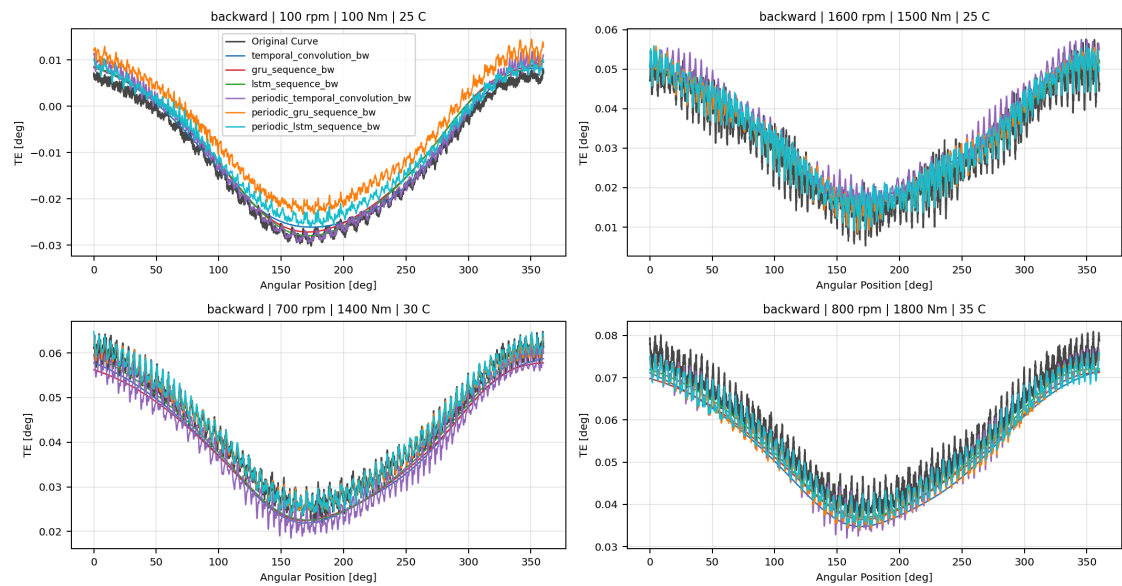


Forward Wave 2.1 Temporal Model Overlay curve-verification comparison

Comparison Gallery - Backward Wave 2.1 Temporal Model Overlay

Included models: `temporal_convolution_bw` , `gru_sequence_bw` , `lstm_sequence_bw` , `periodic_temporal_convolution_bw` , `periodic_gru_sequence_bw` , `periodic_lstm_sequence_bw` .

Backward Wave 2 Temporal Model Overlay

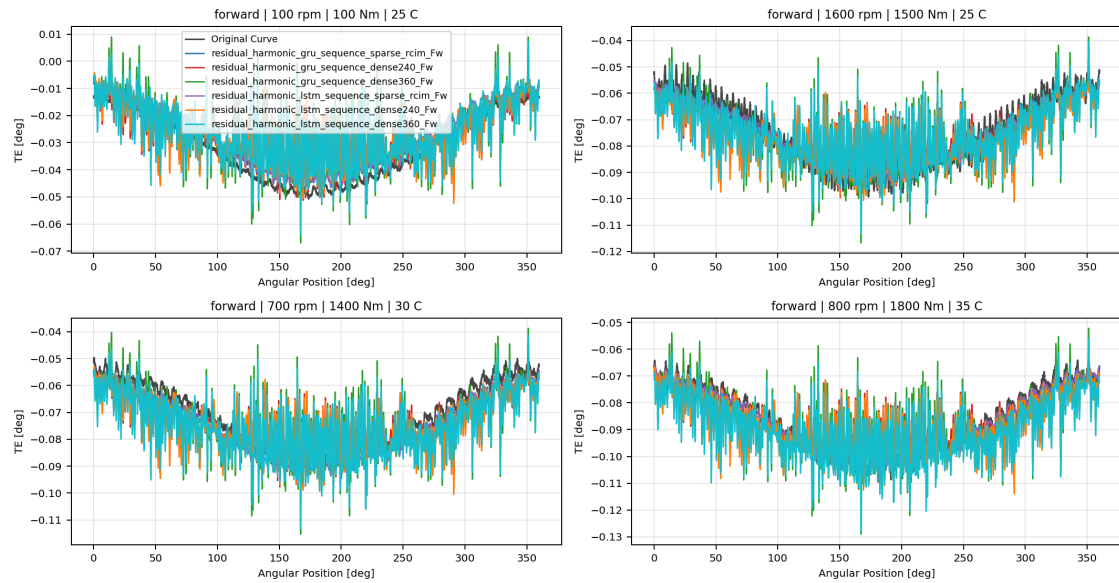


Backward Wave 2.1 Temporal Model Overlay curve-verification comparison

Comparison Gallery - Forward Wave 2.3 Residual Harmonic Temporal Overlay

Included models: `residual_harmonic_gru_sequence_sparse_rcim_Fw` ,
`residual_harmonic_gru_sequence_dense240_Fw` , `residual_harmonic_gru_sequence_dense360_Fw` ,
`residual_harmonic_lstm_sequence_sparse_rcim_Fw` , `residual_harmonic_lstm_sequence_dense240_Fw` ,
`residual_harmonic_lstm_sequence_dense360_Fw` .

Forward Wave 2C Residual Harmonic Temporal Overlay

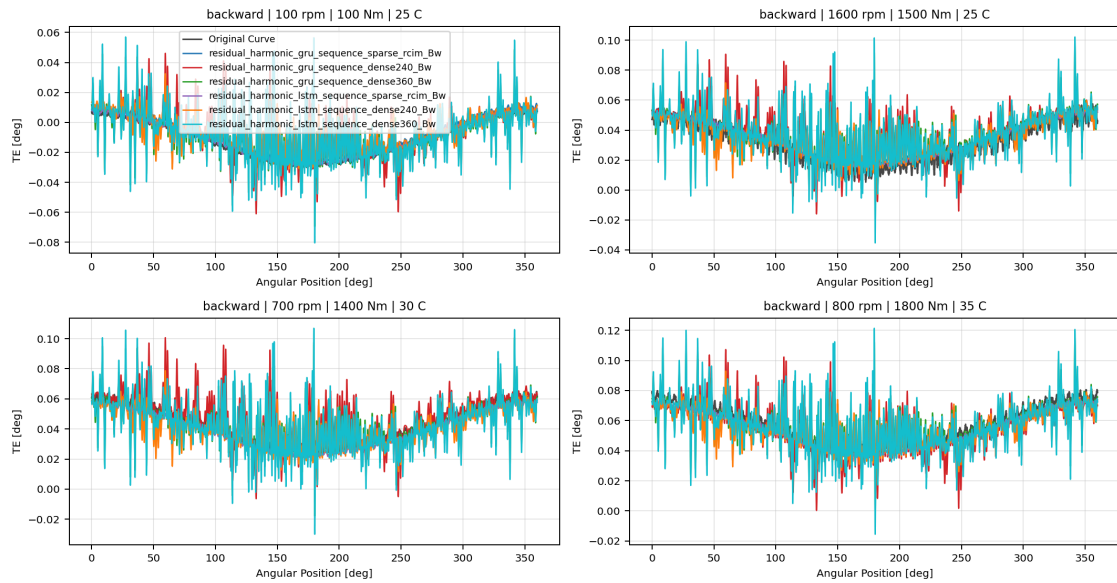


Forward Wave 2.3 Residual Harmonic Temporal Overlay curve-verification comparison

Comparison Gallery - Backward Wave 2.3 Residual Harmonic Temporal Overlay

Included models: `residual_harmonic_gru_sequence_sparse_rcim_Bw` ,
`residual_harmonic_gru_sequence_dense240_Bw` , `residual_harmonic_gru_sequence_dense360_Bw` ,
`residual_harmonic_lstm_sequence_sparse_rcim_Bw` , `residual_harmonic_lstm_sequence_dense240_Bw` ,
`residual_harmonic_lstm_sequence_dense360_Bw` .

Backward Wave 2C Residual Harmonic Temporal Overlay

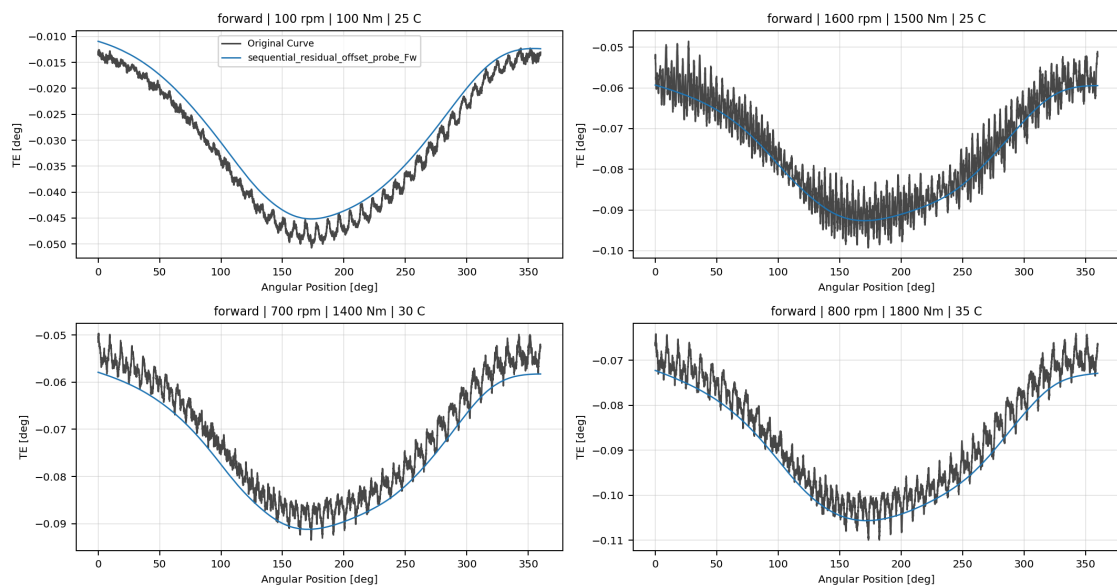


Backward Wave 2.3 Residual Harmonic Temporal Overlay curve-verification comparison

Comparison Gallery - Forward Wave 3.1 Offset-Aware Probe Overlay

Included models: `wave3_1_sequential_residual_offset_probe_fw` .

Forward Track 2F Offset-Aware Probe Overlay

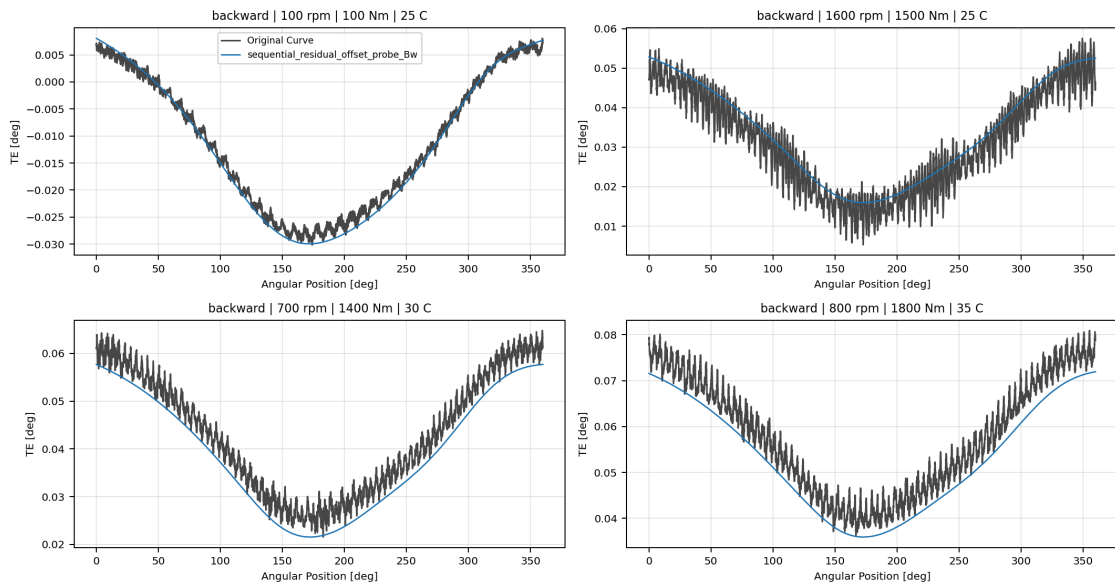


Forward Wave 3.1 Offset-Aware Probe Overlay curve-verification comparison

Comparison Gallery - Backward Wave 3.1 Offset-Aware Probe Overlay

Included models: `wave3_1_sequential_residual_offset_probe_bw` .

Backward Track 2F Offset-Aware Probe Overlay

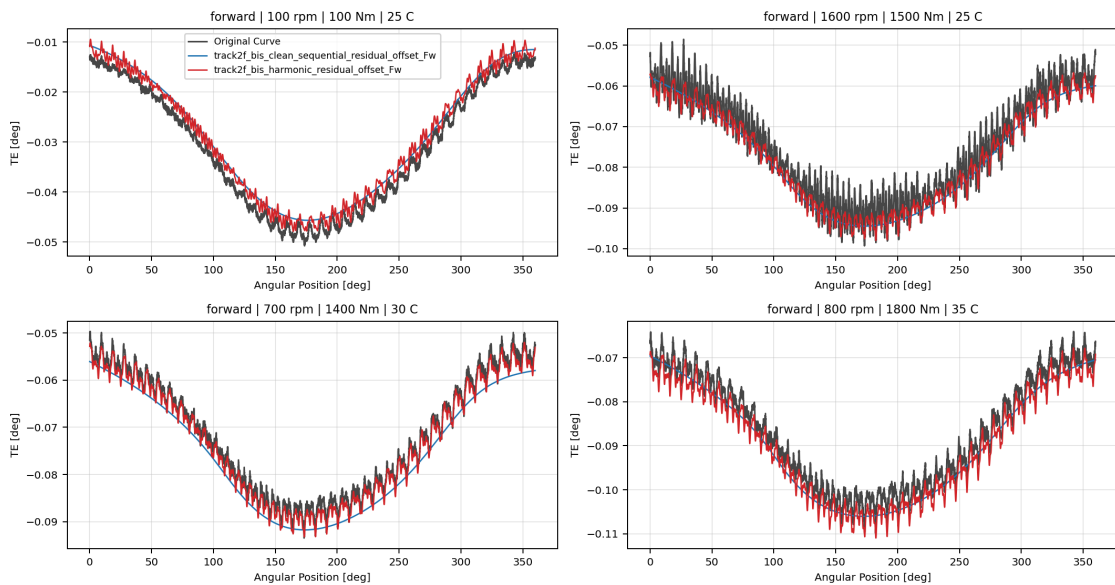


Backward Wave 3.1 Offset-Aware Probe Overlay curve-verification comparison

Comparison Gallery - Forward Wave 3.2 Harmonic-Offset Probe Overlay

Included models: `wave3_2_clean_sequential_residual_offset_fw` , `wave3_2_harmonic_residual_offset_fw` .

Forward Track 2F-Bis Harmonic-Offset Probe Overlay

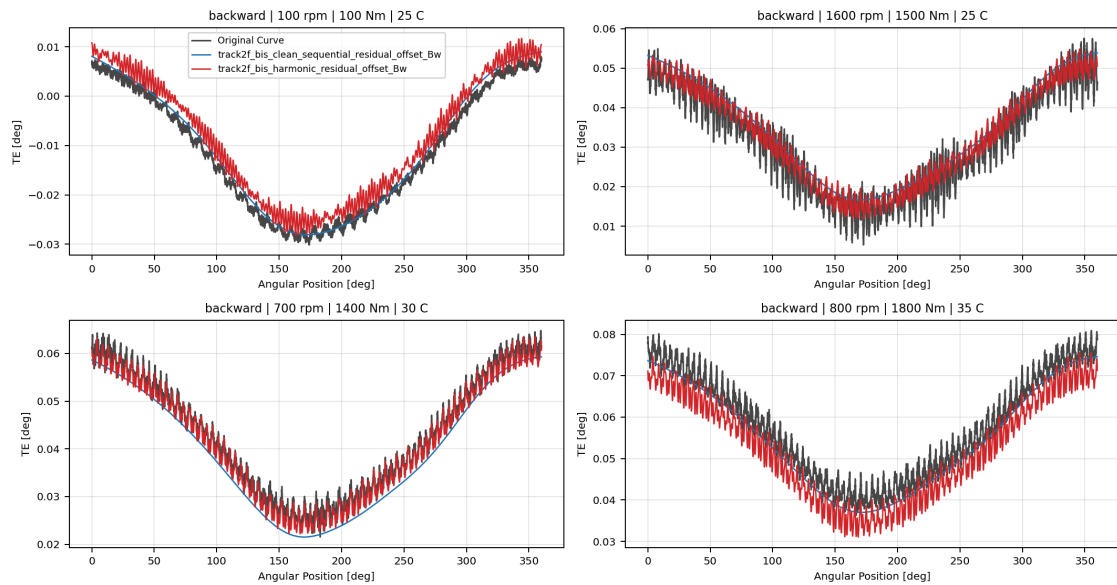


Forward Wave 3.2 Harmonic-Offset Probe Overlay curve-verification comparison

Comparison Gallery - Backward Wave 3.2 Harmonic-Offset Probe Overlay

Included models: `wave3_2_clean_sequential_residual_offset_bw` , `wave3_2_harmonic_residual_offset_bw` .

Backward Track 2F-Bis Harmonic-Offset Probe Overlay

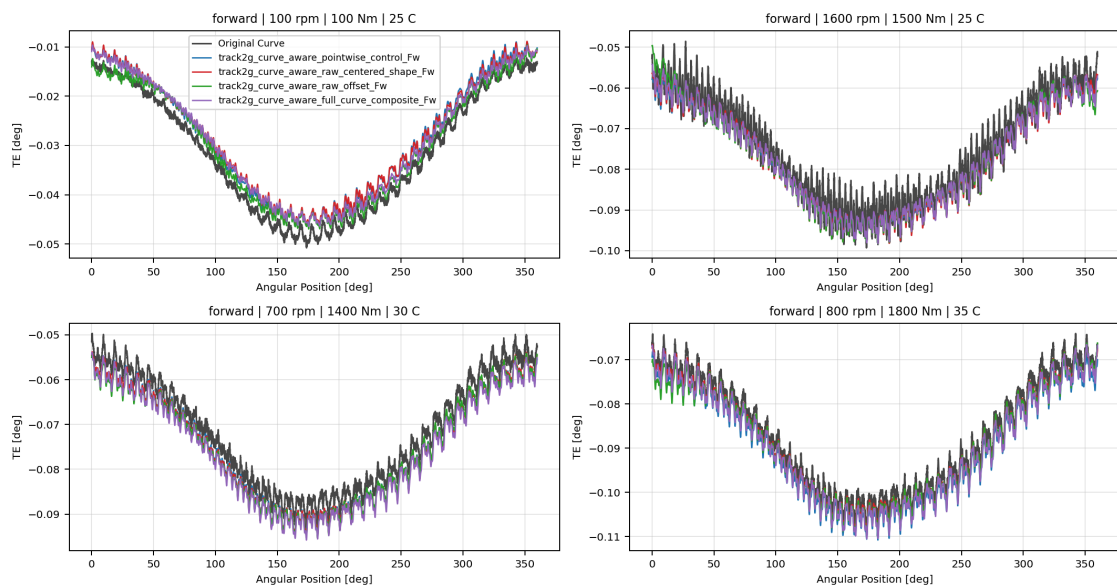


Backward Wave 3.2 Harmonic-Offset Probe Overlay curve-verification comparison

Comparison Gallery - Forward Wave 3.3 Curve-Aware Training Overlay

Included models: `wave3_3_curve_aware_pointwise_control_fw` , `wave3_3_raw_centered_shape_curve_aware_fw` , `wave3_3_raw_offset_curve_aware_fw` , `wave3_3_full_curve_composite_fw` .

Forward Track 2G Curve-Aware Training Overlay

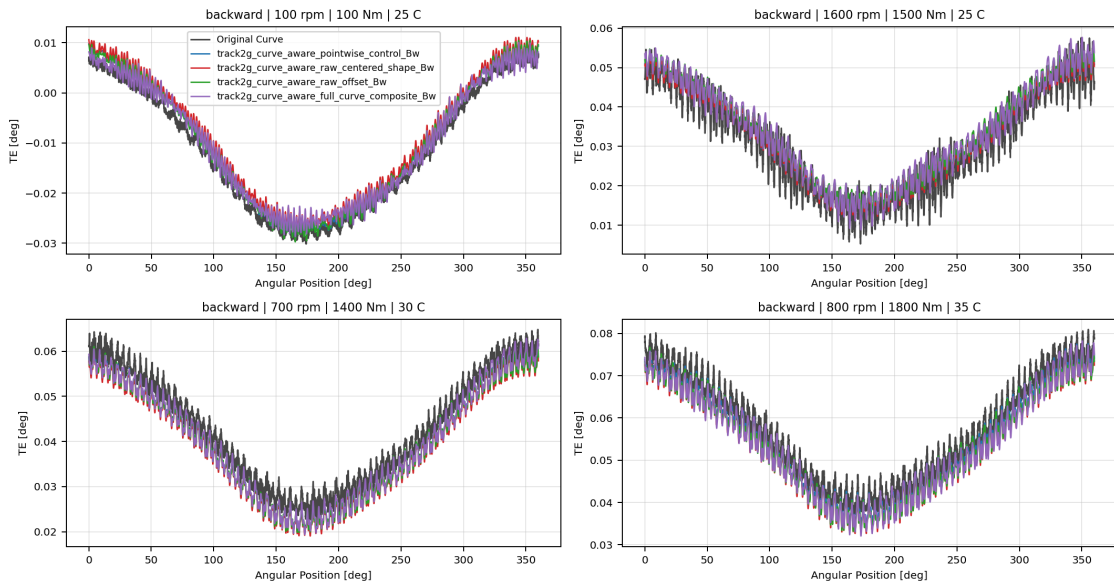


Forward Wave 3.3 Curve-Aware Training Overlay curve-verification comparison

Comparison Gallery - Backward Wave 3.3 Curve-Aware Training Overlay

Included models: `wave3_3_curve_aware_pointwise_control_bw` , `wave3_3_raw_centered_shape_curve_aware_bw` , `wave3_3_raw_offset_curve_aware_bw` , `wave3_3_full_curve_composite_bw` .

Backward Track 2G Curve-Aware Training Overlay

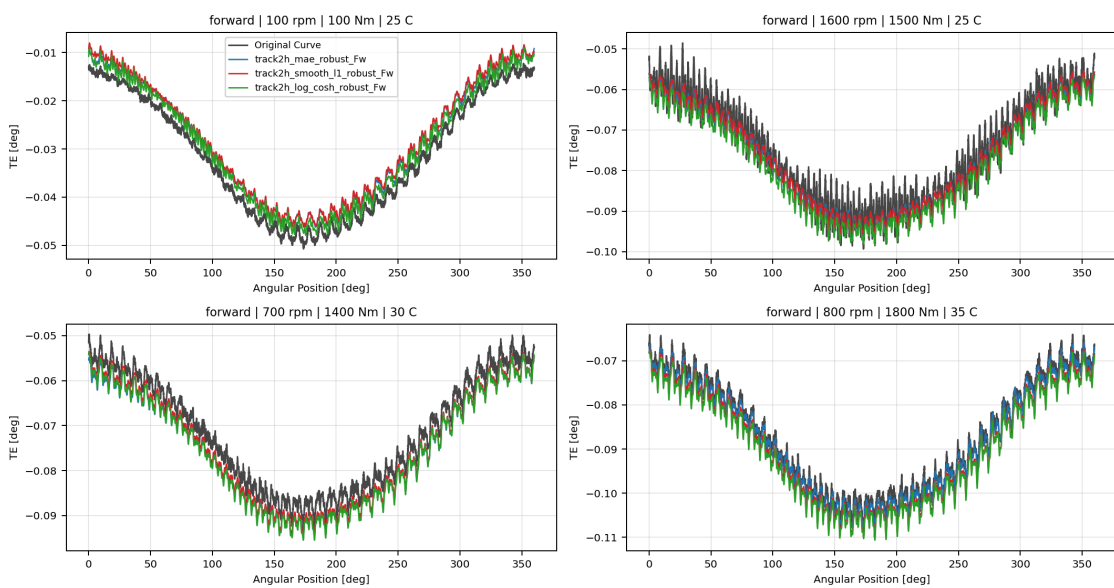


Backward Wave 3.3 Curve-Aware Training Overlay curve-verification comparison

Comparison Gallery - Forward Wave 4.1 Robust-Loss Overlay

Included models: `wave4_1_mae_robust_loss_fw` , `wave4_1_smooth_l1_robust_loss_fw` , `wave4_1_log_cosh_robust_loss_fw` .

Forward Track 2H Robust-Loss Overlay

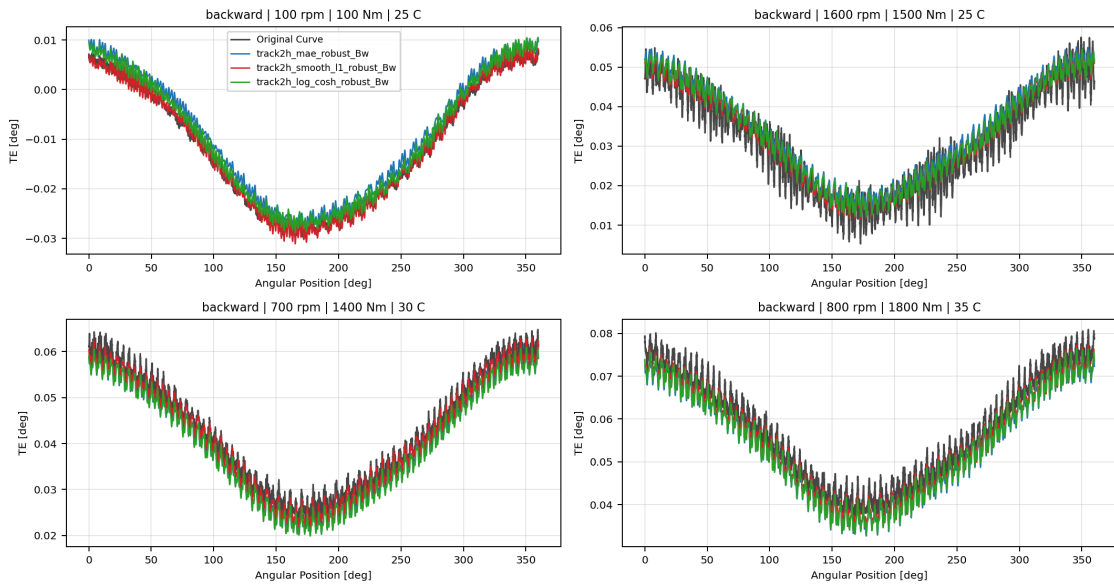


Forward Wave 4.1 Robust-Loss Overlay curve-verification comparison

Comparison Gallery - Backward Wave 4.1 Robust-Loss Overlay

Included models: `wave4_1_mae_robust_loss_bw` , `wave4_1_smooth_l1_robust_loss_bw` ,
`wave4_1_log_cosh_robust_loss_bw` .

Backward Track 2H Robust-Loss Overlay

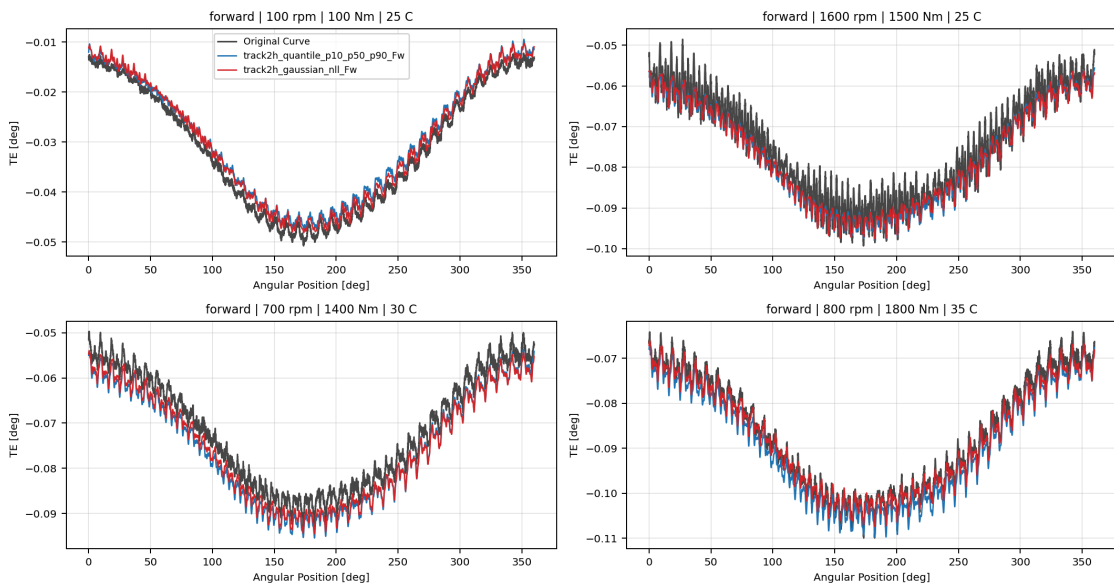


Backward Wave 4.1 Robust-Loss Overlay curve-verification comparison

Comparison Gallery - Forward Wave 4.2 Quantile Probabilistic Overlay

Included models: `wave4_2_quantile_p10_p50_p90_fw` , `wave4_2_gaussian_nll_fw` .

Forward Track 2H Quantile Probabilistic Overlay

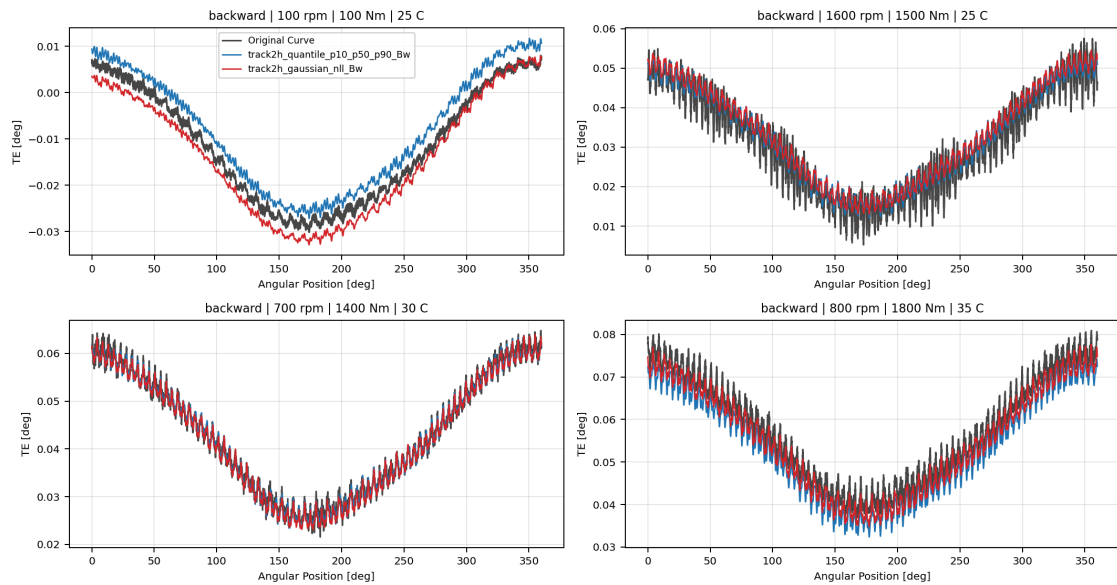


Forward Wave 4.2 Quantile Probabilistic Overlay curve-verification comparison

Comparison Gallery - Backward Wave 4.2 Quantile Probabilistic Overlay

Included models: `wave4_2_quantile_p10_p50_p90_bw` , `wave4_2_gaussian_nll_bw` .

Backward Track 2H Quantile Probabilistic Overlay

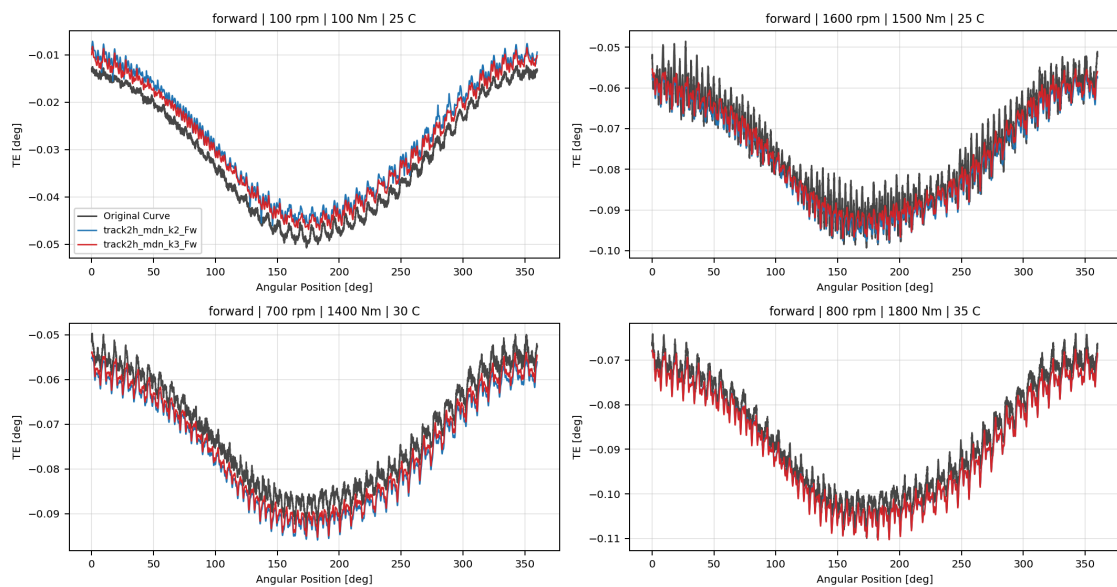


Backward Wave 4.2 Quantile Probabilistic Overlay curve-verification comparison

Comparison Gallery - Forward Wave 4.3 Mixture Density Overlay

Included models: `wave4_3_mixture_density_k2_fw` , `wave4_3_mixture_density_k3_fw` .

Forward Track2h Mixture Density Heads Registry Overlay

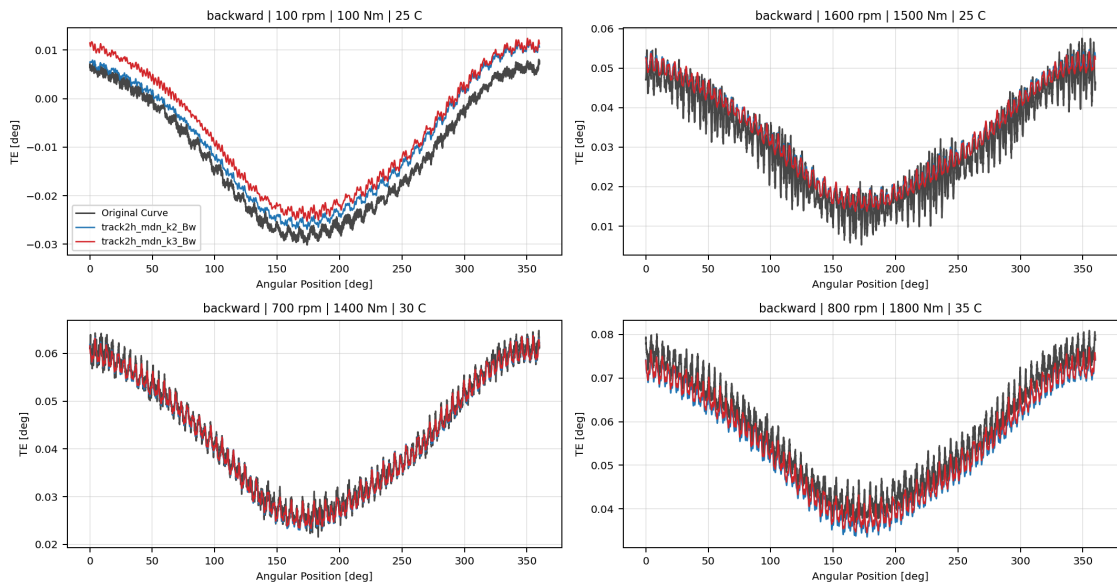


Forward Wave 4.3 Mixture Density Overlay curve-verification comparison

Comparison Gallery - Backward Wave 4.3 Mixture Density Overlay

Included models: `wave4_3_mixture_density_k2_bw` , `wave4_3_mixture_density_k3_bw` .

Backward Track2h Mixture Density Heads Registry Overlay

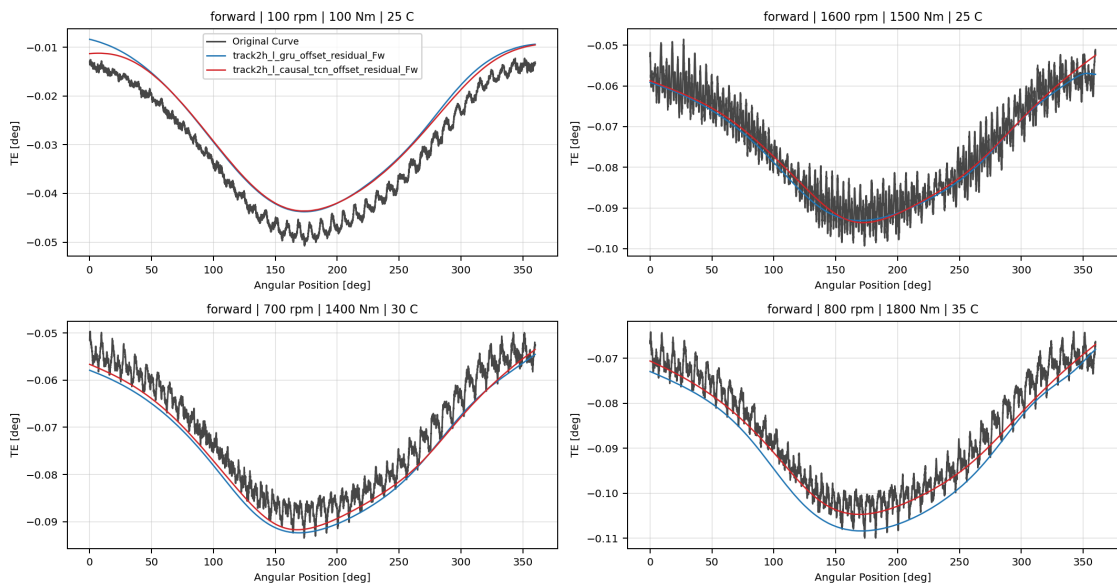


Backward Wave 4.3 Mixture Density Overlay curve-verification comparison

Comparison Gallery - Forward Wave 4.4 Latent State Hysteresis Overlay

Included models: `wave4_4_gru_latent_offset_residual_fw` , `wave4_4_causal_tcn_latent_offset_residual_fw` .

Forward Track2h Latent State Hysteresis Registry Overlay

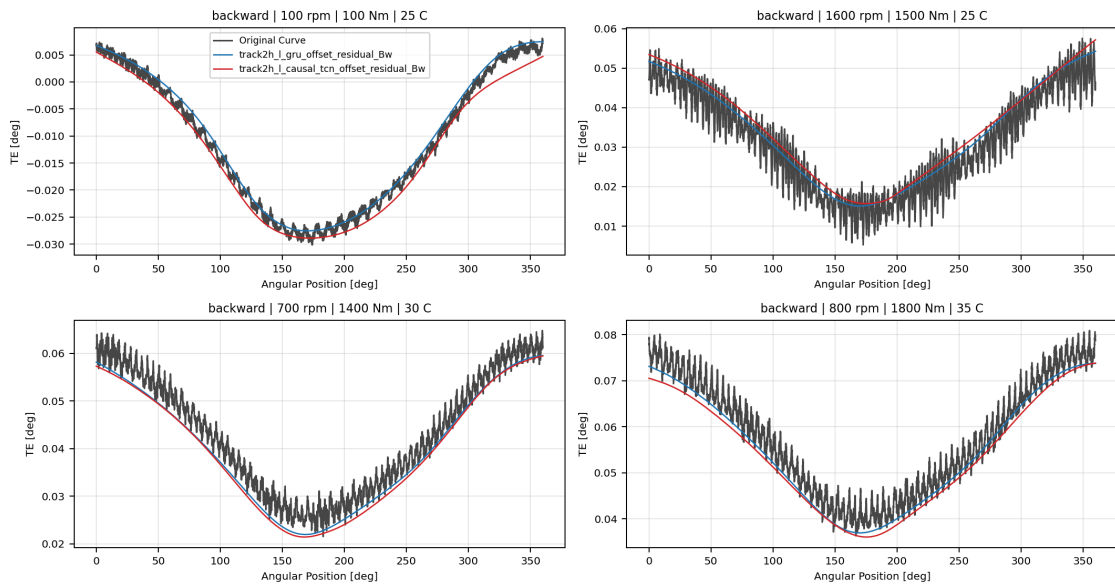


Forward Wave 4.4 Latent State Hysteresis Overlay curve-verification comparison

Comparison Gallery - Backward Wave 4.4 Latent State Hysteresis Overlay

Included models: `wave4_4_gru_latent_offset_residual_bw` , `wave4_4_causal_tcn_latent_offset_residual_bw` .

Backward Track2h Latent State Hysteresis Registry Overlay

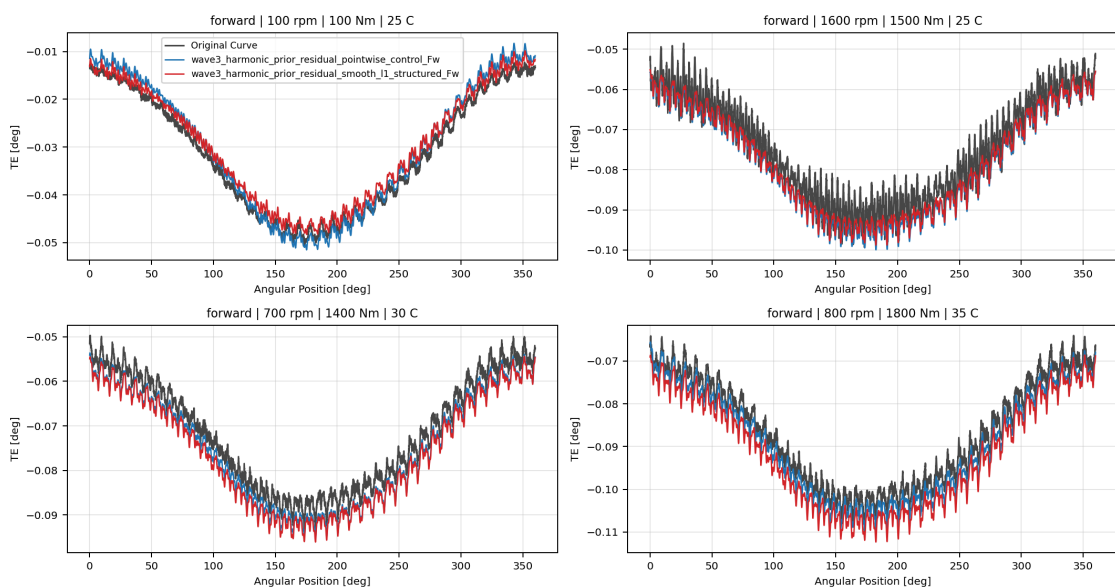


Backward Wave 4.4 Latent State Hysteresis Overlay curve-verification comparison

Comparison Gallery - Forward Wave 5.1 Harmonic Prior Residual Registry Overlay

Included models: `wave5_1_harmonic_prior_pointwise_control_fw` , `wave5_1_harmonic_prior_smooth_l1_structured_fw` .

Forward Wave 3 Harmonic Prior Residual Registry Overlay

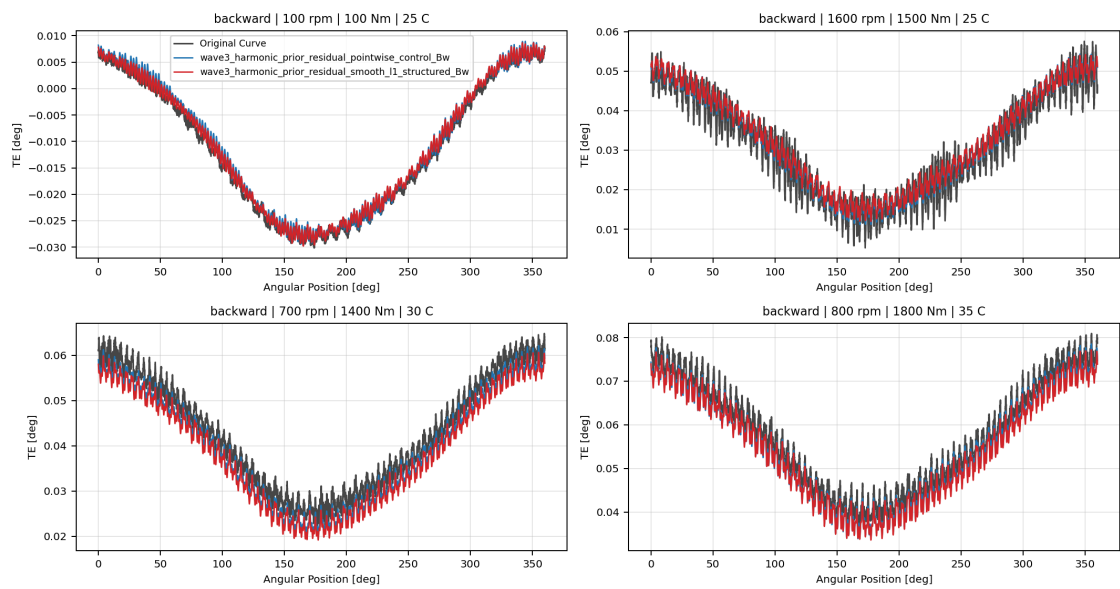


Forward Wave 5.1 Harmonic Prior Residual Registry Overlay curve-verification comparison

Comparison Gallery - Backward Wave 5.1 Harmonic Prior Residual Registry Overlay

Included models: wave5_1_harmonic_prior_pointwise_control_bw ,
wave5_1_harmonic_prior_smooth_l1_structured_bw .

Backward Wave 3 Harmonic Prior Residual Registry Overlay

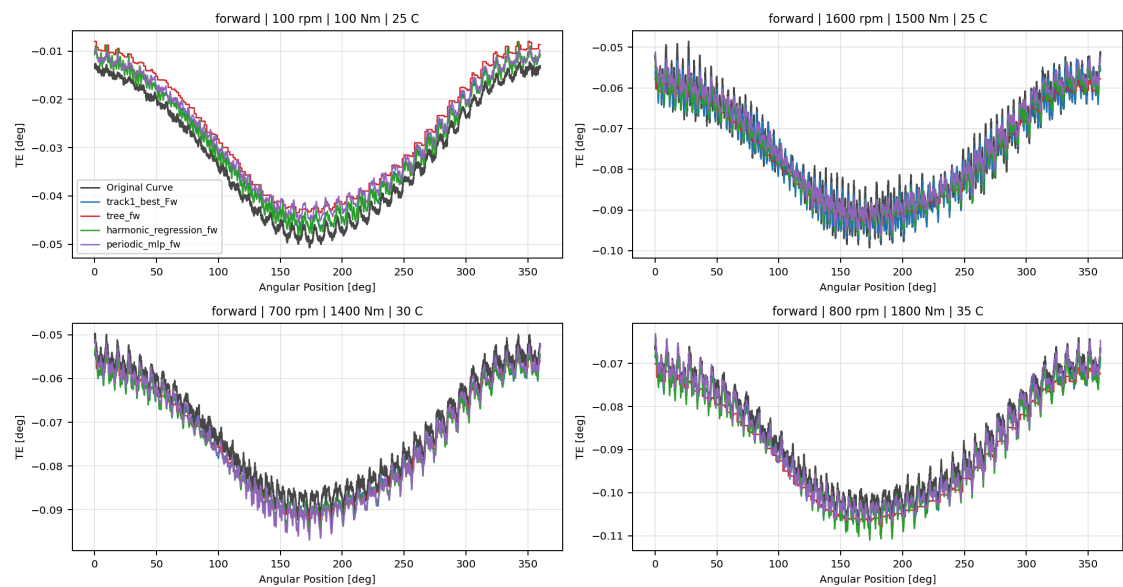


Backward Wave 5.1 Harmonic Prior Residual Registry Overlay curve-verification comparison

Comparison Gallery - Forward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Included models: rcim_model_bank_reproduction_best_fw , tree_fw , harmonic_regression_fw , periodic_mlp_fw .

Forward Track 1 And Screened Wave 1 Overlay

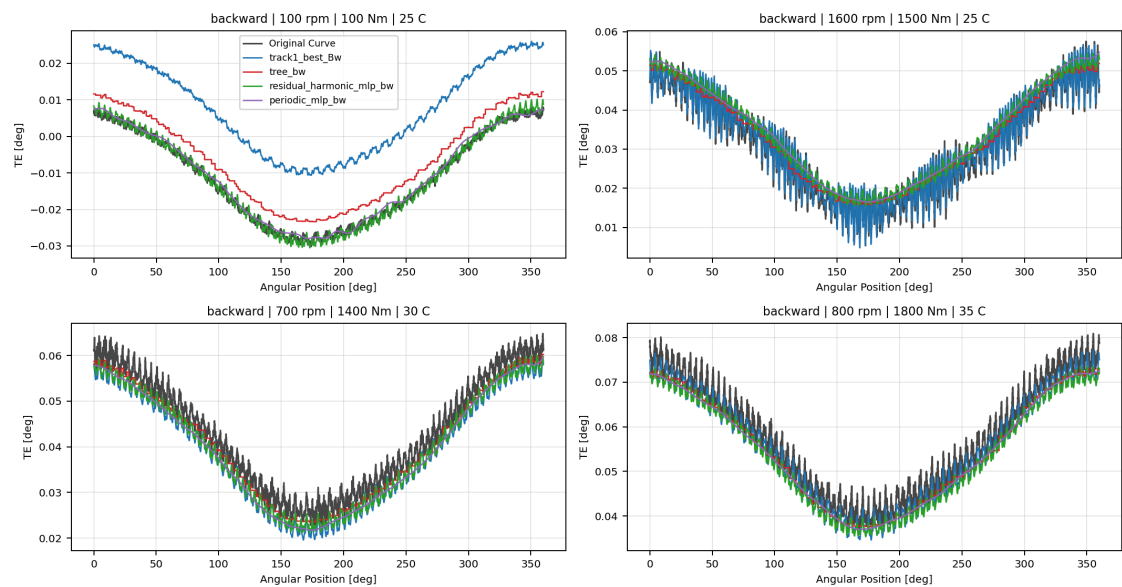


Forward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay curve-verification comparison

Comparison Gallery - Backward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Included models: rcim_model_bank_reproduction_best_bw , tree_bw , residual_harmonic_mlp_bw , periodic_mlp_bw .

Backward Track 1 And Screened Wave 1 Overlay

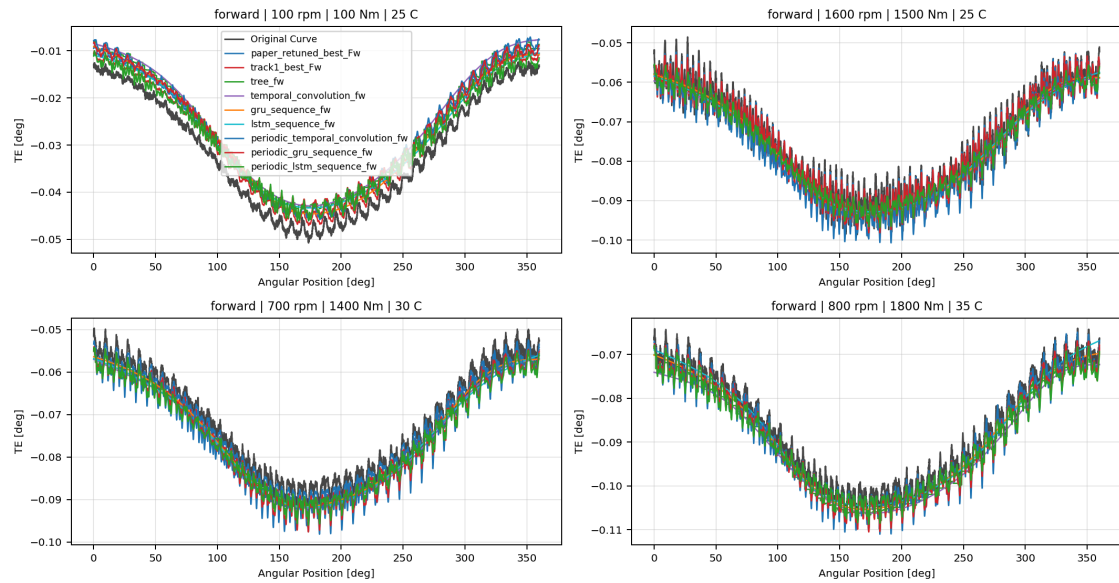


Backward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay curve-verification comparison

Comparison Gallery - Forward Reference Tree And Wave 2.1 Overlay

Included models: `paper_retuned_best_fw`, `rcim_model_bank_reproduction_best_fw`, `tree_fw`, `temporal_convolution_fw`, `gru_sequence_fw`, `lstm_sequence_fw`, `periodic_temporal_convolution_fw`, `periodic_gru_sequence_fw`, `periodic_lstm_sequence_fw`.

Forward Reference Tree And Wave 2 Overlay

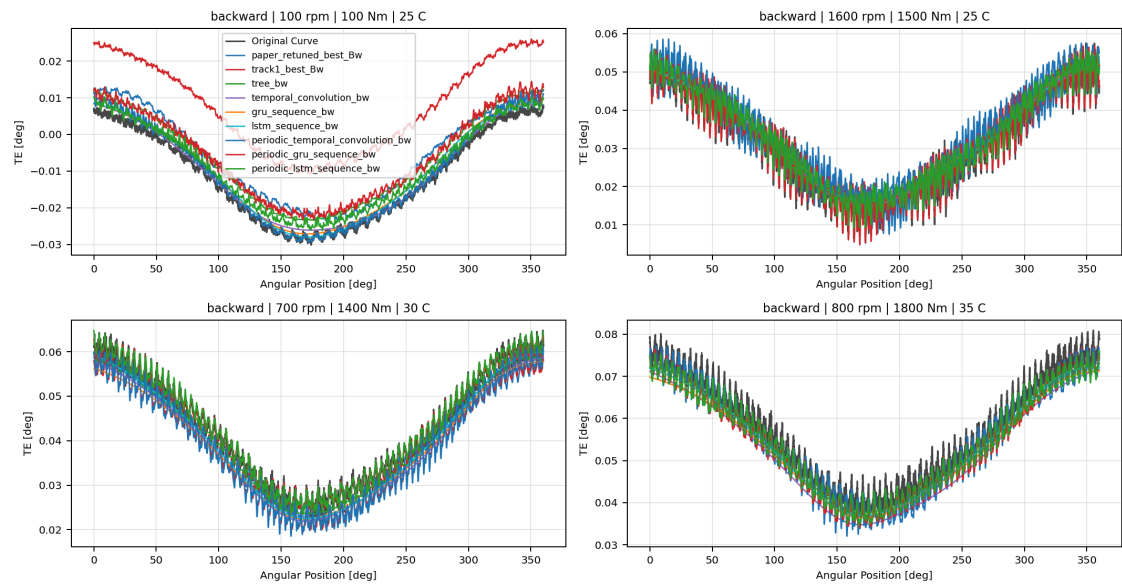


Forward Reference Tree And Wave 2.1 Overlay curve-verification comparison

Comparison Gallery - Backward Reference Tree And Wave 2.1 Overlay

Included models: `paper_retuned_best_Bw` , `rcim_model_bank_reproduction_best_bw` , `tree_bw` , `temporal_convolution_bw` , `gru_sequence_bw` , `lstm_sequence_bw` , `periodic_temporal_convolution_bw` , `periodic_gru_sequence_bw` , `periodic_lstm_sequence_bw` .

Backward Reference Tree And Wave 2 Overlay

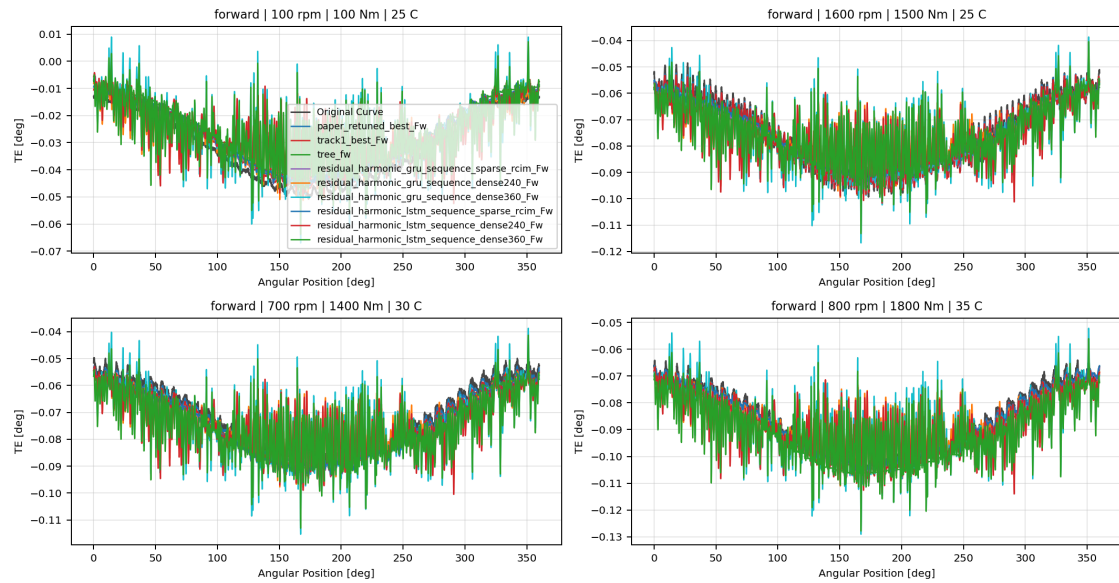


Backward Reference Tree And Wave 2.1 Overlay curve-verification comparison

Comparison Gallery - Forward Reference Tree And Wave 2.3 Overlay

Included models: `paper_retuned_best_Fw`, `rcim_model_bank_reproduction_best_fw`, `tree_fw`, `residual_harmonic_gru_sequence_sparse_rcim_Fw`, `residual_harmonic_gru_sequence_dense240_Fw`, `residual_harmonic_gru_sequence_dense360_Fw`, `residual_harmonic_lstm_sequence_sparse_rcim_Fw`, `residual_harmonic_lstm_sequence_dense240_Fw`, `residual_harmonic_lstm_sequence_dense360_Fw`.

Forward Reference Tree And Wave 2C Overlay

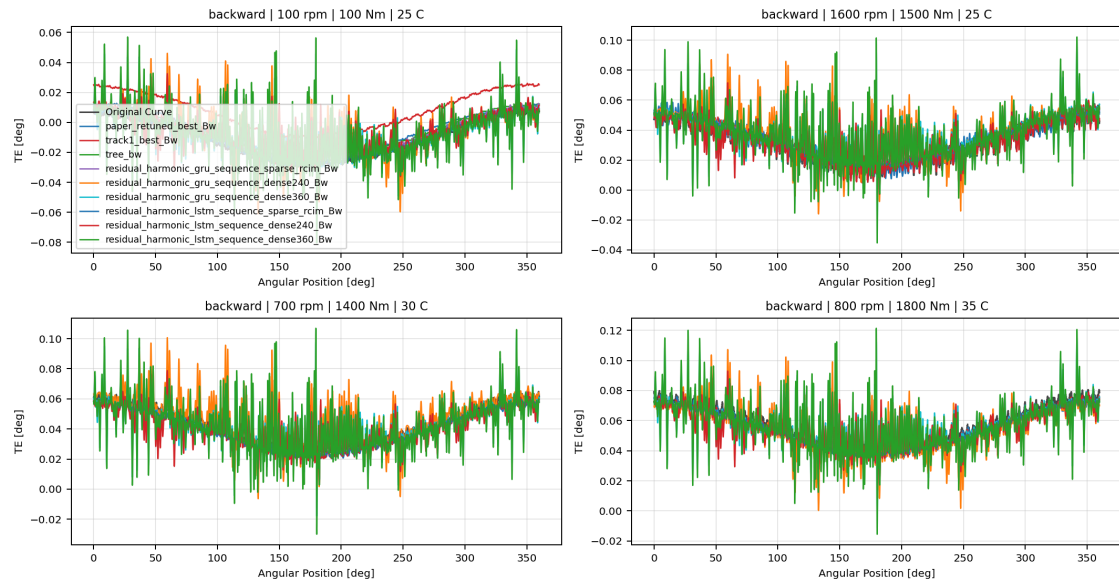


Forward Reference Tree And Wave 2.3 Overlay curve-verification comparison

Comparison Gallery - Backward Reference Tree And Wave 2.3 Overlay

Included models: `paper_retuned_best_Bw`, `rcim_model_bank_reproduction_best_bw`, `tree_bw`, `residual_harmonic_gru_sequence_sparse_rcim_Bw`, `residual_harmonic_gru_sequence_dense240_Bw`, `residual_harmonic_gru_sequence_dense360_Bw`, `residual_harmonic_lstm_sequence_sparse_rcim_Bw`, `residual_harmonic_lstm_sequence_dense240_Bw`, `residual_harmonic_lstm_sequence_dense360_Bw`.

Backward Reference Tree And Wave 2C Overlay

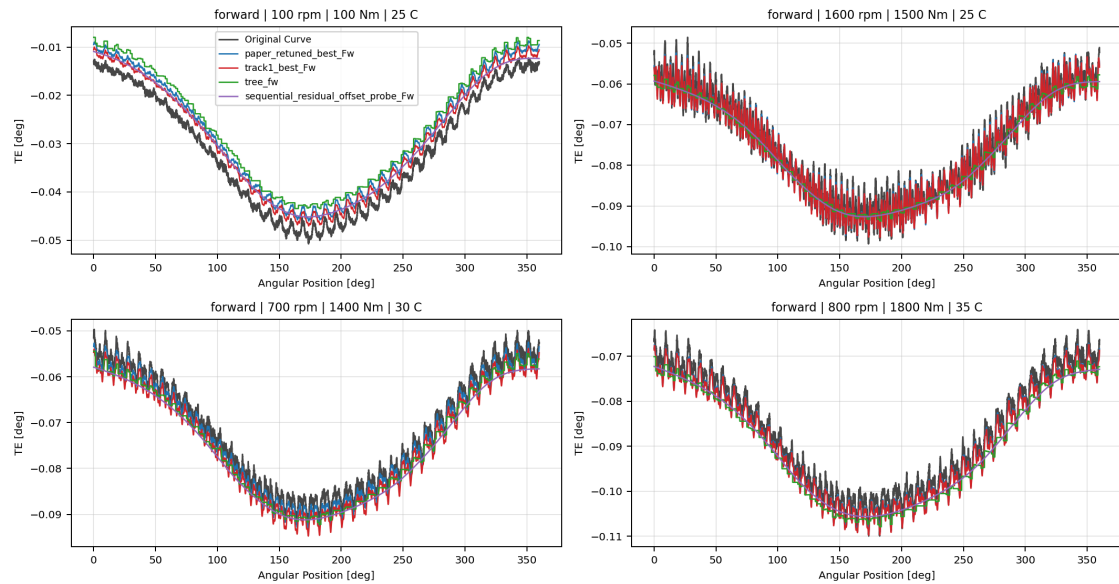


Backward Reference Tree And Wave 2.3 Overlay curve-verification comparison

Comparison Gallery - Forward Reference Tree And Wave 3.1 Overlay

Included models: `paper_retuned_best_Fw` , `rcim_model_bank_reproduction_best_fw` , `tree_fw` , `wave3_1_sequential_residual_offset_probe_fw` .

Forward Reference Tree And Track 2F Overlay

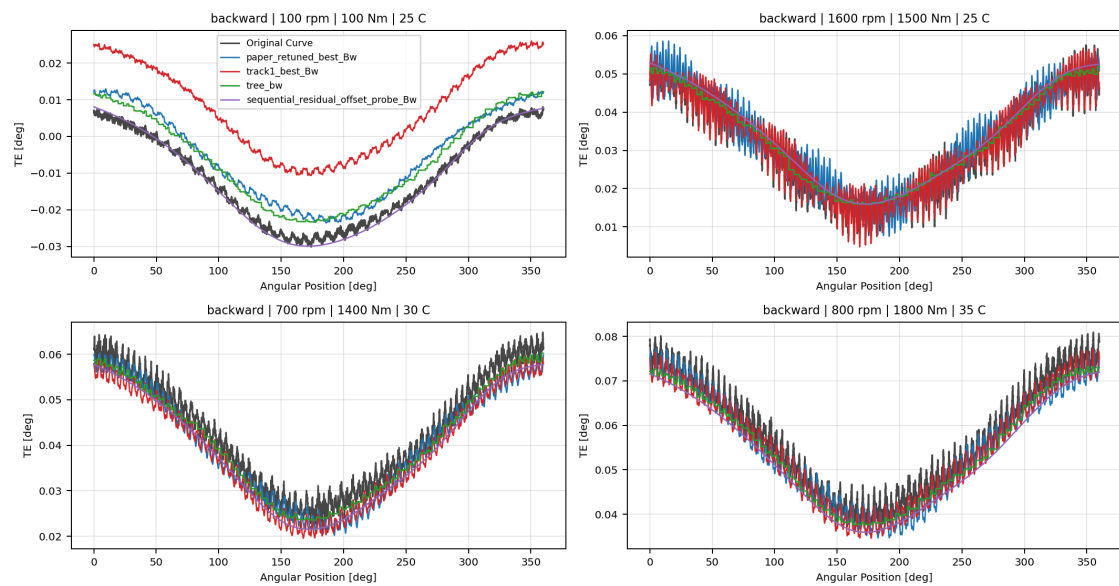


Forward Reference Tree And Wave 3.1 Overlay curve-verification comparison

Comparison Gallery - Backward Reference Tree And Wave 3.1 Overlay

Included models: `paper_retuned_best_Bw` , `rcim_model_bank_reproduction_best_bw` , `tree_bw` , `wave3_1_sequential_residual_offset_probe_bw` .

Backward Reference Tree And Track 2F Overlay

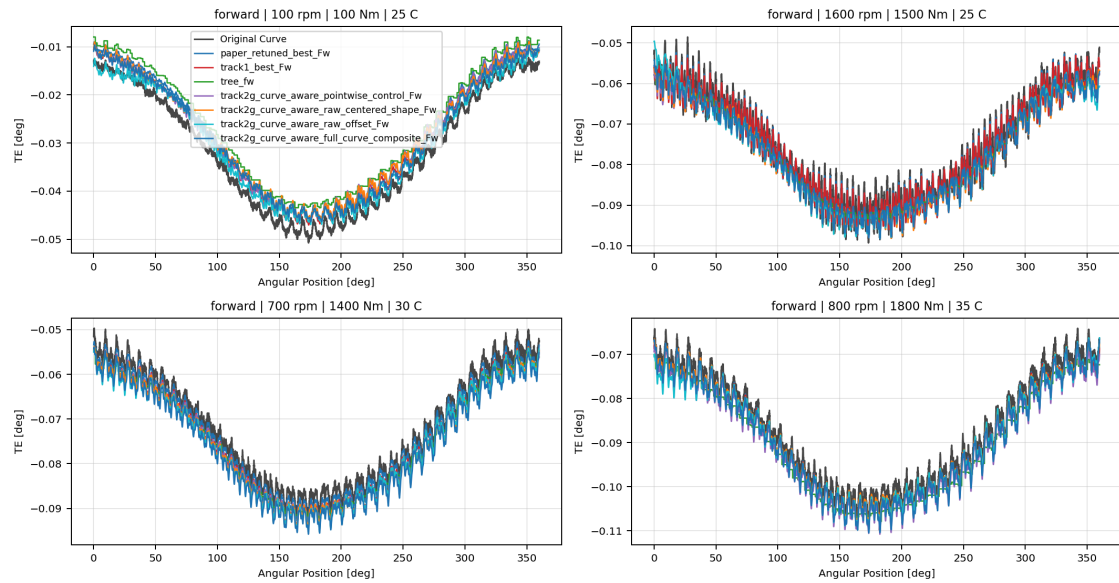


Backward Reference Tree And Wave 3.1 Overlay curve-verification comparison

Comparison Gallery - Forward Reference Tree And Wave 3.3 Overlay

Included models: `paper_retuned_best_fw` , `rcim_model_bank_reproduction_best_fw` , `tree_fw` ,
`wave3_3_curve_aware_pointwise_control_fw` , `wave3_3_raw_centered_shape_curve_aware_fw` ,
`wave3_3_raw_offset_curve_aware_fw` , `wave3_3_full_curve_composite_fw` .

Forward Reference Tree And Track 2G Overlay

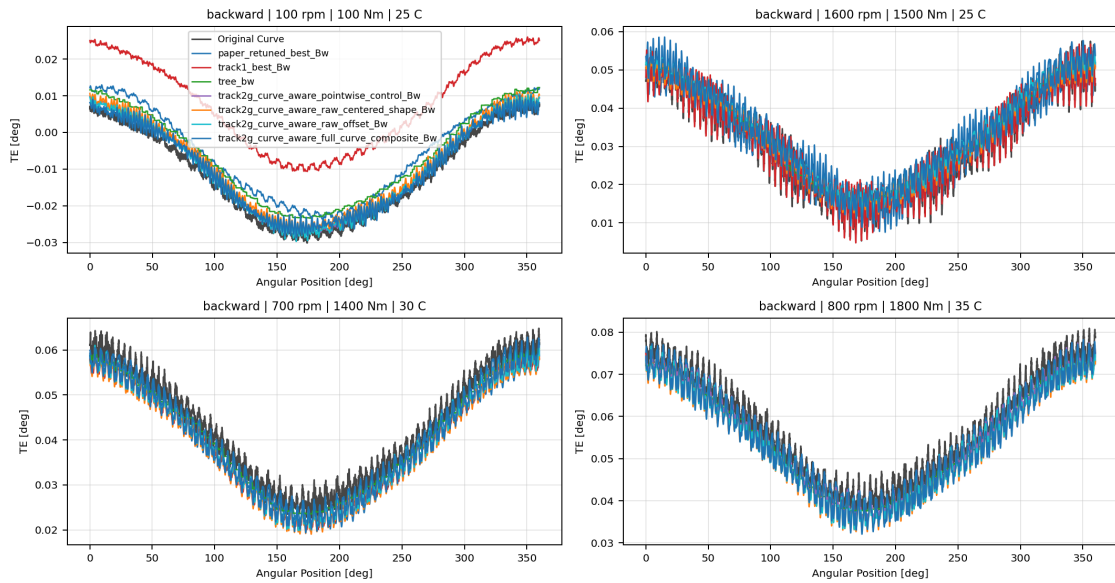


Forward Reference Tree And Wave 3.3 Overlay curve-verification comparison

Comparison Gallery - Backward Reference Tree And Wave 3.3 Overlay

Included models: `paper_retuned_best_Bw` , `rcim_model_bank_reproduction_best_bw` , `tree_bw` ,
`wave3_3_curve_aware_pointwise_control_bw` , `wave3_3_raw_centered_shape_curve_aware_bw` ,
`wave3_3_raw_offset_curve_aware_bw` , `wave3_3_full_curve_composite_bw` .

Backward Reference Tree And Track 2G Overlay

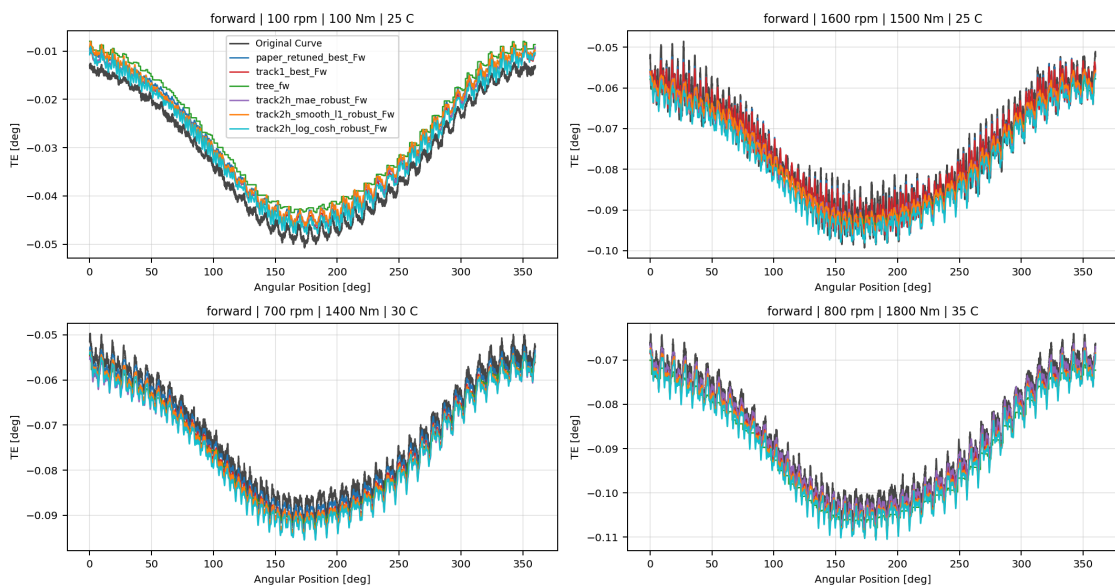


Backward Reference Tree And Wave 3.3 Overlay curve-verification comparison

Comparison Gallery - Forward Reference Tree And Wave 4 series Overlay

Included models: `paper_retuned_best_Fw` , `rcim_model_bank_reproduction_best_fw` , `tree_fw` ,
`wave4_1_mae_robust_loss_fw` , `wave4_1_smooth_l1_robust_loss_fw` , `wave4_1_log_cosh_robust_loss_fw` .

Forward Reference Tree And Track 2H Overlay

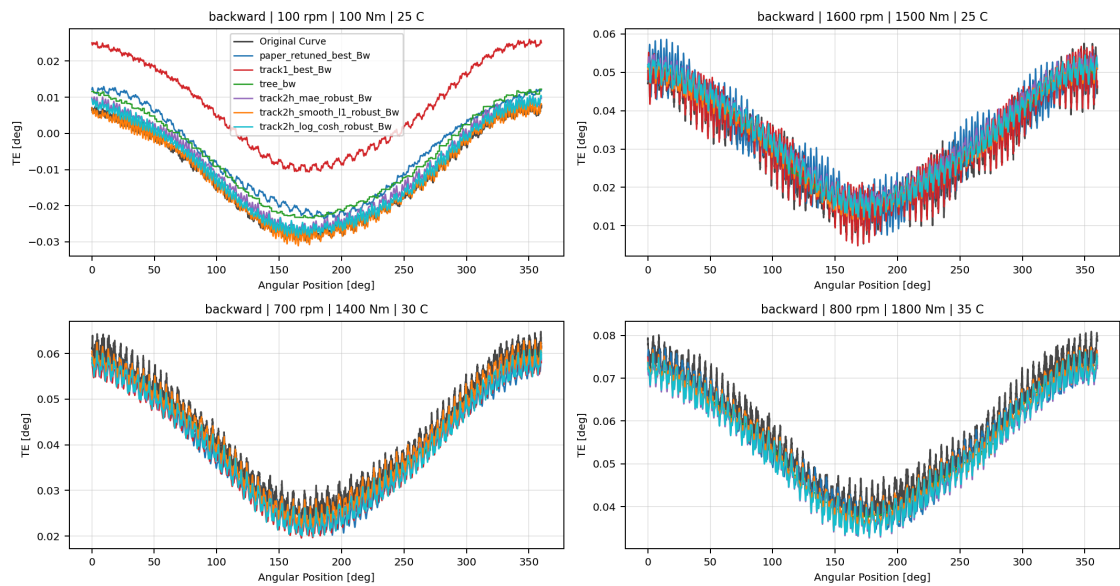


Forward Reference Tree And Wave 4 series Overlay curve-verification comparison

Comparison Gallery - Backward Reference Tree And Wave 4 series Overlay

Included models: `paper_retuned_best_Bw` , `rcim_model_bank_reproduction_best_bw` , `tree_bw` , `wave4_1_mae_robust_loss_bw` , `wave4_1_smooth_l1_robust_loss_bw` , `wave4_1_log_cosh_robust_loss_bw` .

Backward Reference Tree And Track 2H Overlay



Backward Reference Tree And Wave 4 series Overlay curve-verification comparison

Output Artifacts

- output directory: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-06-18-16-54-18__track2_multi_model_curve_comparison_report` ;
- summary YAML: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-06-18-16-54-18__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_summary.yaml` ;
- metrics CSV: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-06-18-16-54-18__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_metrics.csv` ;
- report Markdown: `doc\reports\analysis\track2\multi_model_curve_comparison_report\[2026-06-18]\track2_multi_model_curve_comparison_report.md` .